

PASO DE LOS LIBRES, Argentina

WMO: 872890

Lat: 29.6894S

Lon: 57.1486W

Elev: 70

StdP: 100.49

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.9	4.3	-1.0	3.5	9.0	0.8	4.0	8.3	14.7	14.6	13.3	16.0	1.5	180	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	36.1	24.8	34.8	24.2	33.2	23.7	26.7	32.9	26.0	31.9	25.3	30.6	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.4	30.0	24.3	19.4	29.0	23.8	18.8	28.4	84.5	33.4	81.1	32.0	77.9	30.7	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.6	10.2	DB	-0.3	38.6	1.7	1.3	-1.5	39.5	-2.5	40.3	-3.4	41.0	-4.7	42.0	(4)
			WB	-0.7	28.4	1.8	1.1	-1.9	29.2	-3.0	29.8	-4.0	30.4	-5.3	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	26.4	25.5	23.7	20.5	16.9	14.9	14.2	16.0	17.6	20.4	22.5	25.0
	DBStd	5.55	2.50	2.75	3.07	3.66	3.80	4.55	4.85	4.87	4.17	3.17	2.89	2.85
	HDD10.0	28	0	0	0	0	1	8	13	5	1	0	0	0
	HDD18.3	554	0	1	2	18	75	124	148	106	61	14	4	0
	CDD10.0	3779	507	433	423	317	214	154	144	192	229	324	376	466
	CDD18.3	1262	249	201	167	85	29	20	21	35	39	80	129	208
	CDH23.3	11131	2575	1875	1398	564	135	79	105	274	324	572	1131	2098
	CDH26.7	4597	1220	816	552	186	23	9	17	83	116	183	425	966
Wind	WSAvg	4.2	4.0	3.9	3.7	3.6	3.4	3.7	4.1	4.6	5.2	5.1	4.5	4.2
Precipitation	PrecAvg	1492	133	148	153	159	120	87	83	73	111	163	131	141
	PrecMax	2323	552	377	355	561	322	354	245	188	279	402	527	521
	PrecMin	682	14	27	14	22	5	4	4	2	16	24	9	19
	PrecStd	359	94	86	89	118	83	64	64	51	59	94	101	105
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	37.2	36.4	33.2	29.2	28.0	28.6	31.8	34.0	34.0	35.1	37.4
		MCWB	25.1	25.7	25.0	24.6	22.7	21.6	20.6	21.5	23.4	24.6	23.7	24.5
	2%	DB	36.0	35.1	33.8	31.1	26.8	25.9	26.4	29.1	30.1	30.9	32.8	35.3
		MCWB	24.6	24.4	24.7	24.0	21.0	20.9	20.1	20.8	22.0	22.5	22.6	24.2
	5%	DB	34.2	33.1	31.7	28.8	24.7	23.3	23.8	26.2	27.1	28.6	31.0	33.8
		MCWB	24.0	24.1	23.6	22.6	20.3	19.8	19.3	19.5	20.0	21.6	21.5	23.4
	10%	DB	32.8	31.3	29.8	26.6	22.8	21.3	21.2	23.9	24.2	26.3	29.1	31.9
		MCWB	23.5	23.4	22.8	21.5	19.5	18.6	17.7	18.6	18.7	20.3	21.1	22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.3	26.9	26.0	23.8	22.6	21.7	23.0	24.6	25.7	25.9	27.5
		MCDB	33.6	33.5	33.3	31.0	27.2	26.2	25.6	29.8	31.0	31.9	32.3	33.8
	2%	WB	26.6	26.2	25.9	24.7	22.3	21.4	20.7	21.4	22.6	24.2	24.3	26.5
		MCDB	32.7	32.4	31.4	29.5	25.2	24.6	25.1	27.2	28.5	29.0	29.6	32.2
	5%	WB	25.7	25.4	24.9	23.6	21.0	20.3	19.6	20.2	21.2	22.6	23.3	25.3
		MCDB	31.4	31.0	29.4	27.5	23.5	23.0	23.2	25.5	25.9	26.7	28.2	30.5
	10%	WB	24.9	24.6	23.9	22.5	19.7	19.1	18.4	19.0	19.8	21.4	22.3	24.2
		MCDB	30.0	29.4	28.0	25.3	22.2	21.3	21.3	23.5	23.3	25.1	26.8	29.3
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.5	10.5	9.7	9.2	9.1	9.8	11.0	10.6	9.8	11.1	11.3
		MCDBR	13.5	13.3	13.2	12.3	11.2	10.2	11.5	13.7	14.3	13.1	14.3	14.2
		MCWBR	4.9	4.8	5.3	5.7	6.1	5.9	6.1	6.7	6.9	6.6	6.0	5.5
	5% WB	MCDBR	11.2	10.9	10.8	10.6	8.9	9.1	10.6	12.1	12.4	11.1	11.0	11.6
		MCWBR	5.2	4.8	5.2	5.5	5.4	5.6	6.1	6.5	6.7	6.4	5.8	5.8
Clear-Sky Solar Irradiance	taub	0.408	0.402	0.393	0.394	0.371	0.385	0.385	0.476	0.519	0.458	0.396	0.411	
	taud	2.377	2.398	2.406	2.365	2.406	2.359	2.359	2.332	2.025	1.929	2.169	2.362	2.347
	Ebn at Noon	934	921	896	839	810	767	785	742	763	859	939	934	
	Edh at Noon	130	124	116	109	94	94	100	151	183	154	131	135	
All-Sky Solar Radiation	RadAvg	6.86	6.19	5.35	4.05	2.99	2.53	2.88	3.63	4.46	5.45	6.66	6.95	
	RadStd	0.63	0.39	0.46	0.51	0.31	0.23	0.30	0.32	0.49	0.67	0.77	0.53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.58	+0.61	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	+0.47	+0.54	N/A	N/A	N/A

Nomenclature: See separate page